

Audio services for an asynchronous transfer mode network

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The adoption of asynchronous transfer mode (ATM) as the target technology for the broadband ISDN (B-ISDN) poses many questions for network planners and service providers. This paper addresses the issue of how existing and future audio services are carried on an ATM platform, highlighting the potential and flexibility of ATM over existing switching technologies.

Voice telephony is the mainstay of BT's revenue both nationally and internationally, and so it is very important to migrate this to an ATM network effectively. A technical solution is proposed which addresses many of the doubts which have been expressed for the suitability of ATM for existing plain old telephony services (POTS).

But telephony is just the tip of the iceberg. ATM could offer services such as dial-up CD quality music, wideband voice circuits with very high quality sound, enhanced voice (e.g. for mobile telephones), conferencing facilities, and, of course, multimedia services where video, data and voice are available. These potential services are reviewed along with some details of the coding required.

The paper then proposes some possible network architectures for customers focusing on the technology used and the interworking issues with the existing public switched telephone network (PSTN). The final section investigates quality of service for audio services with particular reference to delay and echo which have to be met within the UK.

Many people see ATM as an overrated pipe dream. Certainly there is a lot of hype, but this paper shows that an ATM platform can deliver the existing services, and is flexible enough to allow BT to carry future applications.

1. Introduction

In 1876 *The Telephone* magazine wrote:

‘When at last this little instrument [the telephone] appeared, consisting, as it does, of parts every one of which is familiar to us, and capable of being put together by an amateur, the disappointment arising from its humble appearance was only partially relieved on finding that it was really able to talk.’

Even accounting for the digital revolution and that phones can no longer be put together from household items by amateurs, the fact remains that the essential quality of voice transmission has changed little in over 100 years.

So, is it time for change? The advent of the B-ISDN using the ATM technology will be able to offer enhanced quality telephony (so called wideband voice) for bi-directional and conferencing calls, and additionally make available other audio services such as CD-quality music and multimedia facilities. Perhaps this is the time to upgrade the

network to enable all customers access to the ‘information superhighway’.

ATM was chosen as the technology to implement the B-ISDN because it offers the possibility of a single network able to support all types of services — audio, data and video [1]. This makes economic sense too, since the current separate networks for voice, data and video can be integrated, which will bring obvious economies of scale.

The ATM protocol can be thought of as a compromise between circuit switching and packet switching. Circuit switching is quite inflexible because it relies on a fixed time slot to send information — ATM does not rely on using the same time slot dedicated to a connection. Rather, the time slot is used whenever the customer has data to send — hence ‘asynchronous’. However, customers do set up connections for calls, so there is some similarity to circuit switched networks.

ATM is based on transmitting all user data in fixed length packets called 'cells'. Each cell has 48 octets available for information, and a 5-octet header for addressing and control. As the ATM protocol does not implement complicated measures to ensure quality (as in X.25), no error protection or flow control on a link-by-link basis is performed. Instead, any such measures are left to higher layer protocols, to ensure that ATM is as simple and flexible as possible. The fixed length cells also allow switching to be performed in hardware which means that high transmission speeds are possible, e.g. 34 Mbit/s, 155 Mbit/s and beyond.

The B-ISDN layered model (see Fig 1) incorporates ATM by using 2 layers. The ATM layer is concerned with cell multiplexing and routing. The layer above carries out all adaptations to higher layer protocols. There are four different ATM adaptation layers (AALs) which are responsible for mapping different types of services to and from the ATM layer. The AAL used depends on the service requirements in terms of the bit rate (variable or constant), connection-oriented or connectionless, and whether a timing relation is required between source and destination.

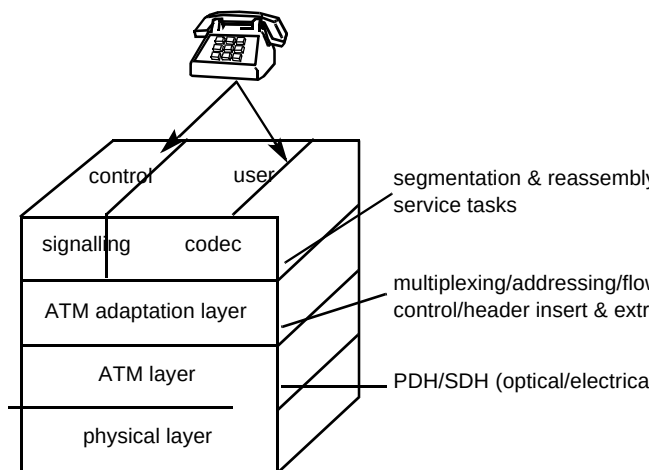


Fig 1 The B-ISDN protocol reference model for audio services.

The paper begins by reviewing the current coding algorithms for audio services showing how ATM can be used as a transport protocol. In section 3, a case study is introduced for the implementation of ATM in the network. Finally, section 4 discusses possible quality of service issues such as delay, echo, and the effects of cell loss on the perceived subjective quality of voice.

2. Audio services

2.1 Introduction

The traffic on today's telecommunications network is predominantly speech. This may well change in the future, as data and video applications are integrated on to a homo-

geneous multiservice network. However, speech communications will still have an important role to play, and are seen as a significant source of revenue in the future.

The speech delivered across today's telecommunications network is based upon 64 kbit/s logarithmic coding. It has a bandwidth of approximately 3.1 kHz, between 300 Hz and 3.4 kHz, and a signal to noise ratio of 38.2 dB. There is an interesting disparity between the quality of speech in telecommunications and audio services in general. Most readers will be familiar with the difference in quality of AM and FM radio, which have a bandwidth of 7 kHz and 15 kHz respectively. Compare this with A-law pulse code modulated (PCM) speech coding, which has a bandwidth of only 3.1 kHz.

The reasons for the relatively low bandwidth of A-law speech coding are that it conveys the essential information, and that existing standards are based upon 64 kbit/s time slots. The advent of ATM and the delivery of broadband services to customers' homes and businesses gives an opportunity to re-evaluate speech transmission quality. If there is a link of many megabits per seconds to customers, it makes no sense to constrain the bandwidth of speech telephony to 3.1 kHz.

2.2 Review of speech coding techniques

Audio information needs to be in the analogue domain to have a direct human interface. It is not possible to understand nor interface directly with digitally encoded information unless this is converted to analogue sound. Therefore, it is necessary to convert analogue information to digital for a digital transmission link, and to reconvert to analogue for the presentation to users. The conversion from an analogue to digital domain, and vice versa, is generally known as 'coding' and 'decoding' respectively. Sometimes, coding also involves the compression of digitally presented information by removing any redundancy or reducing quality.

As telephone systems including transmission media and switching equipment are mostly digital, coding of speech signals is an essential part of a telephone system. The most widely used coding algorithm in telephony is A/μ-Law PCM. The concept of PCM, which was invented and developed by Alec H. Reeves of ITT Laboratories in 1937, is that each sampled voice signal is coded into a combination of pulses for transmission. However, the practical implementation of the PCM system had to wait until the 1960s for the development of transistors and other semiconductor devices.

Apart from the A/μ-Law PCM [2], the International Telecommunications Union (ITU, formerly the CCITT) have also standardised other coding algorithms [3–6]. The International Standards Organisation (ISO) has a coding

algorithm known as the MPEG (moving picture expert group)-audio codec. NICAM (near-instantaneously compressed audio multiplex) is widely used in the broadcasting industry as an audio receiving algorithm of the TV signal. 16-bit linear PCM coding at various sampling rates is used in professional and consumer audio equipment such as CD (compact disc) and DAT (digital audio tape). These algorithms will be explained in more detail later. Apart from these, many proprietary coding algorithms are currently being used between customer premises equipment to increase the number of voice circuits in a digital link. Proprietary audio coding algorithms are also to be incorporated with new compact digital audio equipment such as Philip's digital compact cassette (DCC) and Sony's Mini-Disc in the consumer market. These two devices will make use of compressed digital signals so that the recording media they employ can play for a reasonable duration.

2.2.1 G.711 A/ μ -law PCM

Pulse code modulation (PCM) (see Fig 2) is the most widely established, implemented and applied of all digital coding systems. This is because of its early development and its conceptual simplicity. Most versions of PCM coders are instantaneous, meaning a coding delay of one sample period, and are usually not signal specific. ITU Recommendation G.711 specifies two digital companding systems for 8-bit PCM. A-law companding typifies European PCM practice, and μ -law companding typifies North American and Japanese PCM practices.

Logarithmic companding employing 'A' or ' μ '-law formulae is the dominant coding algorithm as far as the PSTN is concerned. Some voice communications services

over an ATM network will probably follow the same path in using these companding laws. When the ATM network is to interface with the PSTN, conversion to A/ μ -law PCM will be necessary if any other coding algorithms are used in the ATM network.

2.2.2 G.726 ADPCM

This standard is used for the conversion of a 64 kbit/s A/ μ -law PCM channel to and from a lower bit rate channel. The standard covers rates of 40, 32, 24 and 16 kbit/s. The G.726 codec is the second most widely used in telecommunications networks [4]. It was originally conceived in order to increase the capacity of the digital links of long distance networks. This was achieved with the introduction of 32 kbit/s ADPCM technology maintaining a quality comparable to that of 64 kbit/s PCM. The ADPCM codec was the first complex algorithm that the ITU adopted. Once the algorithm was standardised, new areas of application appeared in digital circuit multiplication equipment (DCME), digital cordless telephones and digital speech storage equipment. ADPCM coding at 32 kbit/s was originally covered by standard G.721 [7]. However, this has now been superseded by G.726, which also contains extra coding rates of 40, 24 and 16 kbit/s. The various bit rates are primarily designed for specific applications. The principle application of 40 kbit/s channels is to encode higher bit rate speech-band data (Modem) signals on DCME. 32 kbit/s coding is used for speech calls, while 24 and 16 kbit/s coding schemes are used for overload channels carrying voice on DCME.

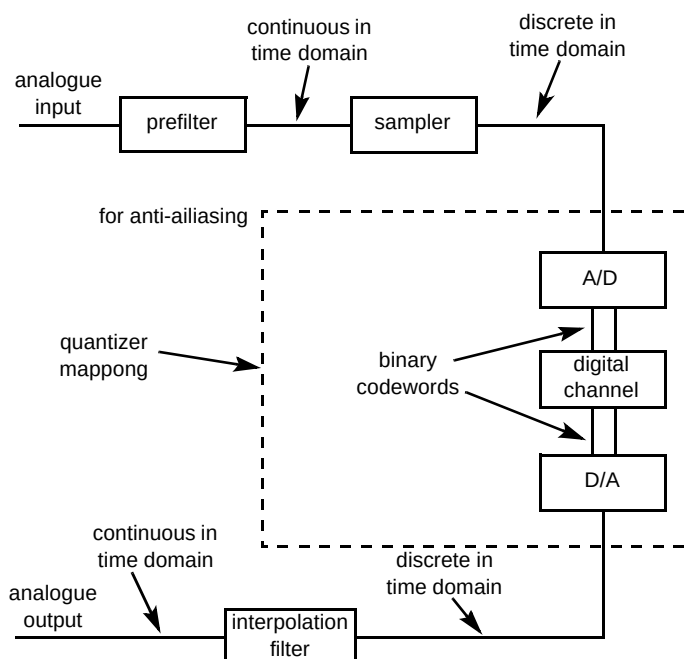


Fig 2 The PCM system.

2.2.3 G.727 embedded ADPCM

G.727 specifies an ADPCM ‘embedded’ coder. Embedded ADPCM algorithms are variable bit rate coding algorithms with the capacity of ‘bit dropping’ outside the encoder and decoder blocks. The ADPCM embedded coder is designed for use in packetised speech systems according to the packetised voice protocol specified in G.764 [8]. The basic idea behind this coding scheme is that the speech signal is encoded as ‘core’ and ‘enhancement’ bits. During network congestion, the enhancement bits can be dropped, giving reduced quality but intelligible speech. This type of coding could be used in ATM networks for congestion control. It would be possible for example to put the core bits in high priority cells, and the enhancement bits in low priority cells. During network congestion the enhancement bits in the low priority cells would be dropped, allowing the core bits to transit the network.

2.2.4 G.722 SB-ADPCM

The ITU standard G.722 for 7 kHz audio is a 64 kbit/s algorithm developed primarily for ISDN teleconferencing and loudspeaker telephony [3]. Because it has the same output rate as A/μ-law PCM at 64 kbit/s, any connection supporting transparent 64 kbit/s digitally can transport a commentary-quality sound programme.

The coding algorithm is based on sub-band adaptive differential pulse code modulation (SB-ADPCM) to reduce an input signal sampled at 16 kHz with 14 bits per sample or 224 kbits to a digital signal for transmission at 8 bits and 8 kHz or 64 kbit/s. The G.722 coding system has three basic modes of operation corresponding to the output bit rates used for 7 kHz audio coding — 64, 56 and 48 kbit/s. The latter two modes allow an auxiliary data channel of 8 and 16 kbit/s respectively to be provided within the 64 kbit/s by making use of bits from the lower sub-band.

2.2.5 G.728 16 kbit/s codec

G.728 specifies a low-delay network-quality speech coding standard at 16 kbit/s for applications in the videophone service, cordless telephone, DCME and packetised speech [6]. The coding algorithm has the combination of high quality with low delay at the relatively low data rate of 16 kbit/s. The algorithm is based upon a technique called ‘codebook excited linear prediction’.

2.2.6 MPEG-audio

A joint working group within the ISO and IEC standardisation bodies initiated the MPEG. The objective was to establish an international standard for encoding of moving pictures and associated audio for storage and retrieval on digital storage media. The main requirement was that the combined bit rate of video and audio should be 1.5 Mbit/s

which is the transfer rate of a CD. The audio codec must work at 64, 96, 128 and 192 kbit/s. This leaves most of the bit rate (out of 1.5 Mbit/s) for the video coding.

The MPEG-audio codec is claimed to be transparent for the coding of 20 kHz audio from a compact disc at the bit rate of 128 kbit/s. This means that there is no perceivable difference between the coded and the original audio. There is much more processing delay with this coder, in the order of tens of milliseconds.

2.2.7 CD audio

The CD audio is recorded digitally with audio information which is converted from analogue to digital format using 16 bits per sample at 44.1 kHz. The resultant bit rate of a single channel audio is 705.6 kbit/s, and for stereo channels the total bit rate is about 1.5 Mbit/s including other information such as the frame header. High quality of decoded audio is achieved without the use of any compression but with the high bit rate. Each sample is encoded with a 16-bits linear quantiser.

2.2.8 NICAM

NICAM was developed by the BBC and uses a near instantaneous companding technique to compress a 14-bit input audio sample into a 10-bit sample one on a 1 ms block basis. The sampling frequency of the input audio is 32 kHz to cover an audio bandwidth of 16 kHz, which is equivalent to FM broadcast. The total bit rate for a NICAM stereo codec is 728 kbit/s including other information such as frame alignment, word and application control bits. This is very widely used in the broadcasting industry for the transmission of high quality audio. Most TV sets currently available with stereo sound have a NICAM decoder.

2.3 ATM convergence

The AAL layer is responsible for the segmentation and reassembly of cells. Each type of AAL uses the 48-byte payload differently, in order to provide the type of service required. AAL1 uses the first byte of the 48-byte user payload to provide a cell sequence number, enabling lost and misinserted cells to be easily detected at the receiving end. It also offers the ability to transmit timing and structure information. AAL0, another AAL which has been advocated to transmit speech signals, uses all 48 bytes of the payload for the speech signal.

For both AAL types, each cell must be filled from the source codec before transmission. This introduces an element of delay into the transmission path. The exact amount of delay depends upon the coding scheme, the AAL, and the fill level used (see section 4.2.2).

When providing constant bit rate (CBR) services over ATM networks, buffering must be provided so that the effects of cell delay variation (CDV) can be overcome. When the source codec sends data it has a constant bit rate. Due to the asynchronous nature of the protocol, the cell stream which arrives at the destination AAL may no longer be constant bit rate. Some cells may become closer together, while others become farther apart. Both these effects are called cell delay variation. The AAL buffer is used to reconstitute the CBR cell stream, so that the destination codec receives the expected constant bit stream as normal (see section 4.2.5).

The standards bodies are currently debating which adaptation layer should be used to convey speech. AAL1 and AAL0 (sometimes known as the null AAL) are the two main contenders. The advocates of AAL0 argue that it gives a 2% bandwidth saving over AAL1. This is because it uses all 48 bytes of the user payload of the cell, rather than 47 bytes for AAL1. However, the proponents of AAL1 argue that the 2% bandwidth saving is not significant, given the superior capabilities of AAL1. These include the following.

- Loss and misinserted cell detection

AAL1 uses the first byte of the user payload to provide a cell sequence number. This provides a mechanism to reliably detect lost and misinserted cells. Lost and misinserted cells occur, for instance, when bit errors in the transmission medium corrupt the addressing information in the ATM header. If left undetected, this would mean that 47 bytes would suddenly be lost or gained in the speech signal (or 48 bytes for AAL0). The resulting phase discontinuity would cause problems with echo cancellation equipment, and also with voice band data.

- Timing and structure information

The adaptive timing method enables end-to-end timing information to be extracted from the cell arrival rate at the receiving end. However, this method has limits to the stability and accuracy of the clock that it can convey. AAL1 offers in addition a more accurate method called synchronous residual time stamping (SRTS). This can be important for some applications, such as the connection of PABXs using ATM

connections. AAL1 also offers the ability to convey structure information from end-to-end. For further details the reader is referred to I.363 [9].

Figure 3 shows the layered model of ATM and its relation to the codec.

3. Network case study

Audio codecs are therefore available to offer enhanced voice and music services which can easily be supported with ATM technology. But what will the network of the future look like? As the network evolves from the existing PSTN to the information superhighway, there are some interesting challenges to overcome. In the interim period, the telephony network may consist of both wideband and narrowband networks.

Figure 4 shows a possible future scenario. Customer A is served by a new ATM-based network. Services are delivered to this customer's home by an APON (ATM over passive optical networks) system. This customer has wideband access, allowing wideband telephony, video on demand, and multimedia computer facilities. Customer B can talk to customer A using the same wideband speech coding scheme, as both customers have wideband access. However, Customer C is on the old PSTN. If this customer is to converse with A, then either both customers need to use a narrowband coder, or some form of transcoder is required, to convert wideband to narrowband signals.

Consider a case where customer A has a wideband coder with 8 kHz bandwidth. Customer C only has a bandwidth of 3.1 kHz on the PSTN. Customer C will hear no difference when talking to A. However, customer A has paid for his wideband service, and expects to have wideband quality. It may therefore be necessary to artificially enhance the quality of the 3.1 kHz call from customer A by speech enhancement techniques.

This highlights one potential interworking problem, but it can be overcome. Such wideband access is clearly a vision of the future which most people would like to have in their home or at work. The differentiators to stimulate

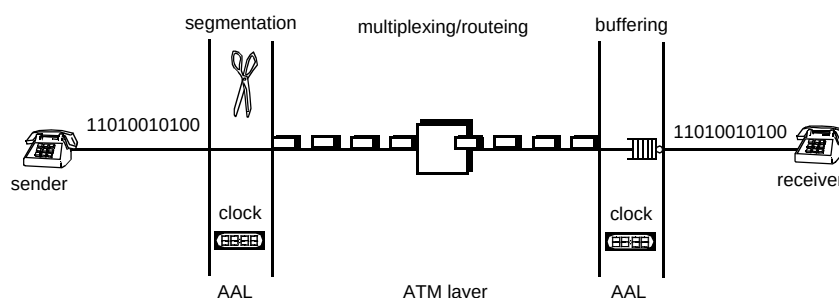


Fig 3 Audio and ATM convergence.

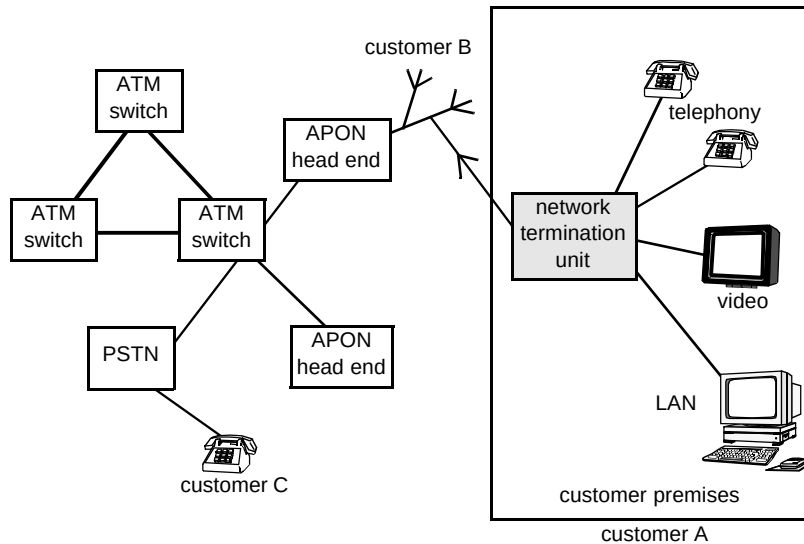


Fig 4 Network case study.

customer demand in these new services are price and quality.

4. Quality of service

As has been seen above, the future telecommunications network is likely to offer more bandwidth to customers. It will therefore be possible to provide a whole new series of higher quality speech and music delivery services.

4.1 Enhancing quality

In the USA, AT&T have started to use quality of service as a major selling point. Echo cancellation is needed on many domestic routes, due to the long distances involved. As part of their echo canceller upgrade program, AT&T have introduced rudimentary signal processing techniques to artificially enhance the quality of their speech calls. Voice signals are subject to automatic level control, plus a gain boost at a frequency of approximately 200 Hz. This is rather similar to pressing the loudness button on a hi-fi to give a more subjectively pleasing sound. This idea, known as Truevoice™, has been the subject of a major advertising

campaign. It is clear that in the future, quality of service will be a major selling point for voice communications services.

The next logical step on the road to quality improvement is to move to mid-band speech coding. For example the G.722 standard can be used to provide 7 kHz speech communications. Further into the future true wide-band speech and even 'surround sound' telephony could be offered to customers. The technology to provide such services is available today.

4.2 How does ATM affect QoS ?

As ATM is a relatively new technology, the way in which it interacts with speech services needs to be examined to ensure that it can offer an acceptable quality of service. Two main areas are examined below — delay and cell loss/misinsertion.

4.2.1 Delay and echo

In the present telephone network, echo can sometimes be heard on international circuits. Figure 5 shows the two principal types of echo. The first is talker echo, where a customer can hear echoes of their own voice. The second

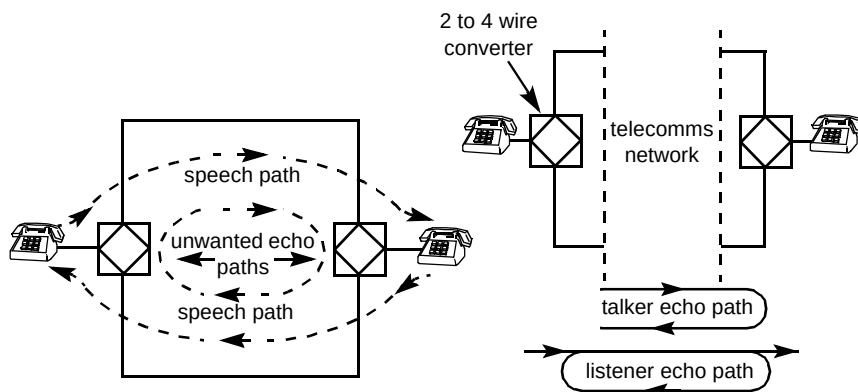


Fig 5 Echo.

form of echo is listener echo, where the listener hears multiple echoes of the speech from the talker. The amount of echo perceived depends upon two variables:

- the amount of delay on the line,
- the amount of reflected energy, which depends upon the echo return loss and the overall loudness rating of the connection [10].

The reflections can arise for example at two-to-four-wire converters in the network. This can also be seen in Fig 5.

In general terms, the amount of echo perceived increases with both delay and loudness. At low delays in the order of a few milliseconds, echo is perceived as side-tone, the normal feedback between microphone and loudspeaker when talking on the phone. Without sidetone the telephone sounds 'dead'. As the delay increases, to around 20 to 30 ms end-to-end, the conversation can sound reverberant, as if it were taking place in a bathroom. Above around 50ms, the delayed feedback sounds more like echo in the true sense of the word.

When sending voice across the ATM network, there will be no change in loudness rating. However, the mechanism and characteristics of the delay are different from the PSTN, so it has to be determined whether echo will be a problem.

4.2.2 Delay guide-lines

One of the more pressing concerns of any network provider will be the stringent delay requirements which have to be adhered to within the UK. Delay standards for the UK network are being discussed by the public network operators [11]. These guide-lines recommend the delay for national PSTN telephone calls with and without echo control facilities.

For national circuits without echo control, the standards state that the end-to-end delay should be less than 23 ms. This is further broken down to allow 5 ms either end for customer networks, which leaves a 13 ms 'budget' for the public network. If the delay guidelines (see Fig 6) are not adhered to, then the subjective quality of the call degrades to an extent where the customer may complain. The next

section describes the main sources of delay in the public network.

4.2.3 Delay in an ATM system

Delays in ATM-based speech transmission have three causes:

- cell packetisation delay,
- transmission delay,
- cell playout buffer delay.

The cell packetisation delay is simply the time needed to fill an ATM cell with speech samples before transmission. Using AAL1 this equates to 47 samples of speech. The actual time to fill the cell depends upon the bit rate of the speech coding scheme. For example, with A-law encoded speech at 64 kbit/s, the delay is given by:

$$\text{Delay} = \frac{8 \text{ bits/sample}}{64 \text{ kbit/s}} \times 47 = 5.875 \text{ ms.}$$

Table 1 shows the packetisation delay introduced for some of the codecs described previously. Clearly, the delay is higher at low bit rates.

The second element of delay in an ATM network is due to transmission and switching, which is in the order of a few milliseconds. This is much less than the existing PSTN, as ATM avoids the need for 'multiplexing mountains'. This is where, for example, a 64 kbit/s speech circuit is first multiplexed on to a 2 Mbit/s channel, then to 34 Mbit/s and then up to 140 Mbit/s. The circuit is then demultiplexed though 34 and 2 Mbit/s again before it reaches its final destination. Each multiplexing stage adds extra delay to the circuit. ATM, with its flat structure, avoids this problem.

The cell playout buffer delay normally adds about 1ms for a 64 kbit/s channel. The total delay for a cell with 64 kbit/s coding is therefore around 7 ms, plus the transmission delay. Therefore for a 'completely ATM' call between Manchester and London (approximately 400 km), the total one way delay would be of the order 9 ms, assuming the transmission delay is 5 s per km. This is within the budget of 13 ms. For most national calls the delay

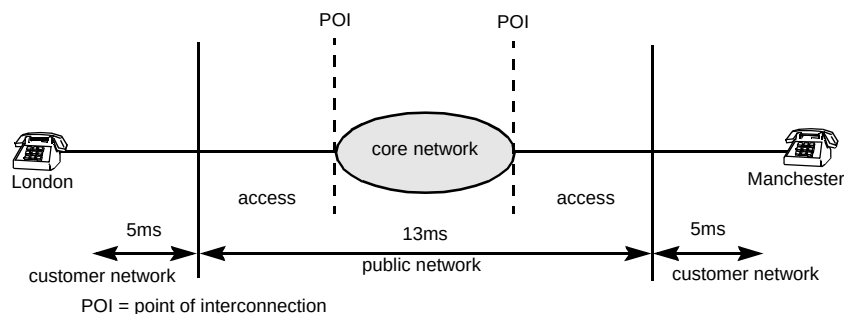


Fig 6 Delay requirements for end-to-end voice.

Table 1 Packetisation delay introduced for some codecs.

Codec	CD	NICAM	PCM	ADPCM	G.722
Bit rate	1.5 Mbit/s	728 kbit/s	64 kbit/s	32 kbit/s	16 kbit/s
Packetisation delay	0.251 ms	0.517 ms	5.875 ms	11.75 ms	23.5 ms

will be acceptable, and in some cases better than the PSTN. If, however, voice services are run across a combination of ATM and PSTN networks, the total delay may exceed the 13 ms allowed for national circuits.

4.2.4 An all-ATM network

There are potentially three methods to cope with the delay inherent in the ATM system. The first is to partially fill the ATM cells. This means that the adaptation layer does not wait for a cell to be completely filled before transmission. The second is to use composite cells consisting of speech samples from many different sources. However, this makes switching much more complicated. The third method is to use echo-cancellation at the customer premises equipment (CPE), when the end-to-end delay is greater than the permitted amount.

In an 'all-ATM' network, no echo cancellation will be required for national 64 kbit/s calls. Echo cancellation will be needed for many international calls, because of the physical distances involved. The question arises of where to put the echo cancellation function. At present, echo cancellation is performed at an international switching centre (ISC). This is a remote centre which handles only international calls which are routed out of the UK via satellite or undersea cables. It is easy to identify speech circuits at the ISC, as they are contained within 64 kbit/s slots of physical 2 Mbit/s circuits. Some speech circuits are passed through echo cancellation equipment. However, consider a scenario in the future, when ATM traffic is

switched internationally (depending on the destination). Cells from many different types of service such as speech, data and video will be contained within a single virtual path. It will be extremely difficult to determine which are speech cells and which are video or data cells. How are the speech cells isolated for echo cancellation? Multimedia applications pose even more of a problem as speech, video and data may be encoded within a single cell by customer CPE.

This leads to the conclusion that the best place to put echo cancellation functionality is within the customer's CPE, at the very edges of the network. This has the added advantage that placing the echo cancellation function near to the source of the echo makes the echo canceller's job much easier, as the echo path is shorter.

Instead of providing echo cancellation, partially filled cells could be used instead. A problem here is that partially filled cells waste bandwidth. In some applications this is acceptable, but not in others. Say, for example, a 16 kbit/s LD-CELP encoder is used to conserve bandwidth on an ATM-based wide area network. If the cells are then only partially filled, it defeats the object of using a speech compression algorithm. In this case, bandwidth is saved by efficient speech coding, but wasted by partially filling the cells.

From this it can be concluded that echo cancellation at the customers CPE is the best technical solution.

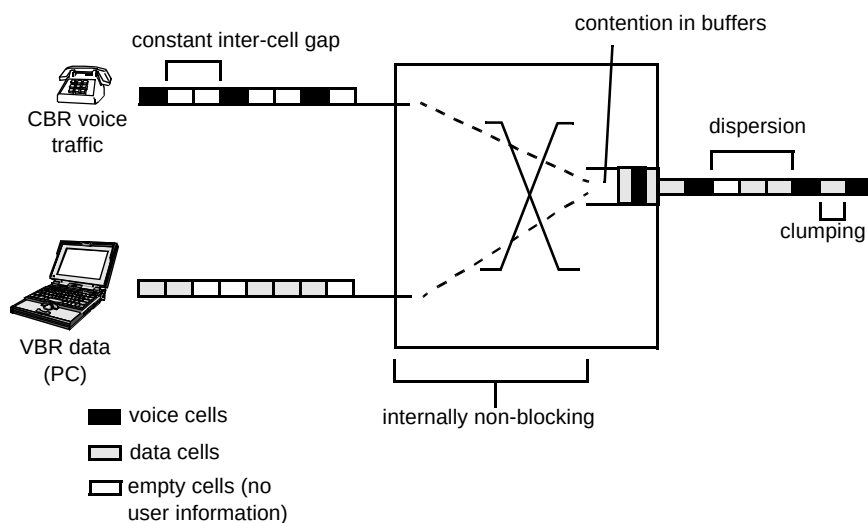


Fig 7 ATM multiplexing and CVD.

4.2.5 Cell delay variation

Most ATM multiplexers are internally non-blocking with contention occurring at the output buffers of the switch (see Fig 7). Due to the slotted nature of the transport system, the contention means that there is some degree of delay variation for ATM cells. For example, for PCM systems the bit stream from the codec is packetised into cells which are sent to the network with a constant inter-cell time; the cell stream which arrives at the receive end is distorted in the network, so that there is no longer a constant inter-cell gap. Some cells come closer together (clumping), while others move further apart (dispersion). The playout buffer in AAL1 needs to be dimensioned so that the constant inter-cell gap can be reconstructed for the decoder at the receive side of the connection. This is a question of predicting the worst case CDV which can occur in a CBR cell stream when it is multiplexed with other traffic. Many studies have been carried out which have tried to predict the worst-case CDV experienced by a CBR cell stream after a number of ATM multiplexers [12, 13]). The models are in the most part very conservative when compared to a simulation.

Figure 8 shows the cumulative function for the delay from the simulation model for a 10-node network plotted in microseconds for a 155 Mbit/s link. Table 2 defines the maximum CDV for a national 64 kbit/s voice connection with the following assumptions:

- there is no echo cancellation,
- 64 kbit/s PCM-coded speech is considered,
- fully filled cells are used.

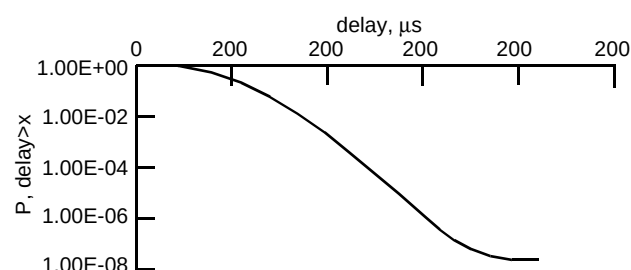


Fig 8 Cell delay variation for ten nodes from the simulation model.

To enable a single figure to be used for all telephony connections the maximum CDV has been agreed within BT to be 1 ms. This means that for each 64 kbit/s call, the AAL1 buffer waits for a maximum of 1 ms after the expected time for the next ATM cell to arrive. If the cell

does not arrive within this interval then it is considered lost by the AAL. The procedure for dealing with lost cells is considered in the next section.

4.3 Cell loss and misinsertion

Cells can be lost in an ATM network for a number of reasons. They can be lost due to random bit errors occurring in the addressing information of the ATM header during transmission, discarded during overloading of the ATM switching fabric, or discarded due to late arrival at the AAL buffer because of excessive cell delay variation (CDV). A lost cell is equivalent to a loss of 376 bits of information in the speech signal.

The effects of losing cells vary with coding type and cell replacement strategy used. AAL1 uses 1 byte of the 48-byte ATM payload to provide a 3-bit sequence number, plus CRC protection. This sequence number can be used to detect lost or misinserted cells. Lost cells must be replaced to ensure bit count integrity, and misinserted cells must be removed.

Currently, lost cells are replaced by all 1s. This means that each cell lost is replaced by 376 binary 1s. For speech signals, this may not be the best strategy to maintain quality of service. Work has been carried out at BT Laboratories to discover the subjective effect of lost cells on an A-law encoded speech signal. Pre-recorded speech signals were A-law encoded and sent across a simulated ATM network. The cell loss ratio of the network was varied, and the output speech recorded. This was repeated for two different cell replacement strategies, 'all 1s' and then 'repeat the last cell'. The speech signal was then decoded, and recorded. Subjective testing was then performed on the recorded speech.

A summary of the results is as follows:

- a lost cell can cause an audible click, when replaced with 47 bytes of all 1s,
- this is objectionable, especially at high cell loss ratios,
- replacing a lost cell with the previous received cell gives much better subjective quality for voice.

In general, losing a cell at lower bit rates causes greater problems, as more of the speech signal is lost. For example, at 64 kbit/s, a cell corresponds to 5.875 ms. This increases to 47 ms at a coding rate of 8 kbit/s. Different cell

Table 2 Maximum CDV for a national 64 kbit/s voice connection.

Interface*	Load	Nodes	10^{-7}	10^{-8}	10^{-9}
34 Mbit/s	0.8	5	1 ms	1.07 ms	1.21 ms
155 Mbit/s	0.93	10	0.63 ms	0.7 ms	0.84 ms

* These results are from simulation studies carried out at BT Laboratories.

replacement strategies may be needed for other coding schemes, or when carrying voice band data.

Linear predictors used in many of the compression codecs rely on the incoming samples to update their algorithms. If a cell is lost this will certainly have an effect on the next prediction, possibly sending the prediction in completely the wrong direction, which may take some time for the algorithm to correct completely. Therefore, a single lost cell could have an effect on a number of subsequent speech samples. The mechanism for overcoming this effect remains for further study.

4.4 Statistical multiplexing

ATM is essentially a fast-packet protocol, so there is some potential for statistical gain for both constant and variable bit rate traffic streams. Most of the current voice codecs were designed to accommodate an intrinsically synchronous network protocol with fixed time slots. Therefore, their output stream has a constant bit rate even though speech is, in itself, bursty in nature. Future codecs may be developed to produce a VBR cell stream, and this would allow statistical multiplexing of voice calls. However, there are some early indications that statistical multiplexing may be possible for constant bit rate voice telephony calls when mixed with other more bursty traffic.

In fact, due to the relatively low bit rate of telephone calls in the context of ATM transmission rates, statistical multiplexing of many voice calls with other data and video traffic could well yield significant increases in revenue because more calls can make use of the available bandwidth.

Early ATM networks will rely on allocating network resources based on the peak rate of the connection. This would apply to both constant bit rate (e.g. voice telephony), and variable bit rate connections (such as data and some video). Peak rate allocation is fine for CBR connections, but wastes bandwidth for VBR connections whose bandwidth requirements may be much less than the peak for most of the duration of a call. Some data applications for example could have a peak requirement of 10 Mbit/s, but most of the time might be transmitting at around 1 Mbit/s or less.

To free up this wasted bandwidth and take advantage of the potential for statistical multiplexing, future ATM networks will use intelligent connection acceptance control (CAC) algorithms to make a decision on whether or not a connection can be accepted on to the network. Many of these algorithms are based on the concept of an effective bandwidth which is a statistical estimate of the bandwidth requirement over the duration of a call. The effective bandwidth is normally calculated using mathematical models [14, 15], and is defined somewhere between the peak bit rate and the mean bit rate of a call. For voice telephony calls the effective bandwidth is just the peak rate since they have a constant bit rate. Voice calls are fairly

robust to cell loss, which means that other VBR calls could be multiplexed with the voice cells and statistical multiplexing achieved. Figure 9 shows the connection acceptance boundary for two call classes using the concept of the effective bandwidth. As the number of VBR video conferencing calls increases (x axis), the potential for statistical multiplexing increases compared to peak rate allocation (the dotted line). This shows that statistical gain is both possible and achievable when mixing CBR voice calls with VBR traffic. The QoS here is defined by a cell loss probability of 10^{-7} .

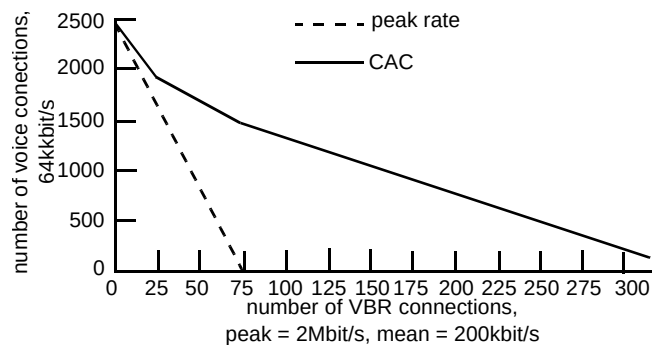


Fig 9 Peak rate versus effective bandwidth CAC.

4.5 The future of speech coding

The speech coding algorithms described earlier were originally developed for one or more specific transmission/storage media. For example, A/ μ -law PCM was developed at the time to allow transmission of speech over a digital link economically with reasonable quality. The G.722 SB-ADPCM 7 kHz codec was developed to take advantage of the end-to-end digital link of an ISDN connection. This also gave the opportunity to increase the quality and bandwidth of the codec.

The speech coders mentioned in section 2 are essentially fixed rate, having been designed for traditional PCM systems. They remove some of the redundant information in speech to produce a constant bit rate stream. However, to take full advantage of the time-varying nature of audio information and of the flexible bandwidth of an ATM network, a variable bit rate coding algorithm is required. Variable bit rate coding can achieve the required output quality while adjusting the codec's bit rate in accordance with the speech signal. For example voiced sounds like 'o' have a narrower bandwidth than fricatives such as 'f', and therefore need fewer bits to encode them.

Most complex coding algorithms currently available take advantage of the redundancy and correlation between the current and past samples to reduce the transmission bit rate. This can cause distortions generated by missing cells to propagate much longer than the duration of the missing cells. New coding techniques based on the cell boundaries of ATM may be able to overcome this problem.

More radical approaches in the development of coding algorithms are required to exploit the full potential of ATM. The requirements for the new audio coding algorithms in ATM should have variable bit rates, layered coding hierarchy, speech activity detection (SAD) and echo cancellation. The variable bit rate takes advantage of the bursty nature of speech. The layered coding approach will be more suitable for the variable capacity of transmission media and the diverse capabilities of decoding terminals. The lowest layer is decoded for the minimum quality audio output while higher quality is achieved with the decoding of more layers. This is the most efficient way to connect different terminals in applications such as conference calls. The incorporation of SAD and echo cancellation makes more economical use of transmission capacity and the cost saving of hardware respectively.

A future use of variable bit rate coding in ATM networks is for congestion control. It would be possible, for instance, for the ATM network to signal to the coder that congestion was taking place. The coder would then reduce its output bit rate. As calls finish, and more network bandwidth becomes available, the network could then signal this fact to the codecs, allowing higher bit rate and higher quality communications once more.

Therefore congestion management and the additional possibility of statistical multiplexing, should mean that the production of VBR codecs is a realistic and cost-effective method for the transmission of voice and other audio services in the future.

5. Conclusions

Since the adoption of ATM as the target technology for the B-ISDN, there has been a lot of interest in multimedia types of application, at the expense of current and future audio services. Voice telephony will continue to be the major revenue earner for telecommunications operators into the next century, and therefore it is very important that these services can be supported by ATM as well.

This paper has shown that ATM is capable of supporting voice telephony, and indeed could enhance the quality of service. Due to the greater bandwidth available other services, such as NICAM and CD-quality audio, may now be supported on the network. They could be offered as stand-alone services, e.g. for enhanced TV, or a network 'Juke-box', or as part of a multimedia application.

Clearly, ATM will enable many residential and business customers the opportunity to experience the information superhighway at first hand. It is the responsibility of the network provider to roll out the technology, and then to price these services competitively so that people are encouraged to use them. It is hoped that the paper has shown that there is great scope for BT to offer both new and existing audio services. Perhaps in 100 years time, this

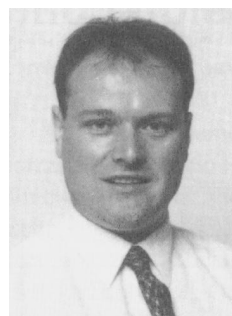
decade will be seen as another turning point in telecommunications history.

Acknowledgements

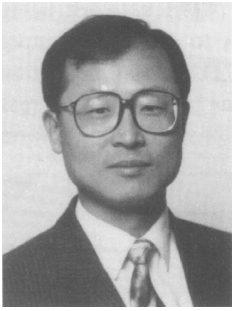
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